Engineering Mechanics: Free Body Diagrams

Quick notes (sample)

1. What is a free body diagram?

A free body diagram (FBD) isolates one body and shows every external force and moment acting on it.

2. Steps to draw one

- Choose the body and imagine it separated from its surroundings.
- Draw the weight W = m g, acting downward through the centre of gravity.
- Add every contact force: normal reactions, friction, tension, applied loads.
- Mark the angles and distances you will need.

3. Equilibrium equations (2D)

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Sum of forces in x = 0
Sum of forces in y = 0
Sum of moments about any point = 0
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4. Worked example

A 20 kg block rests on a level floor with coefficient of static friction 0.3. A horizontal force of 50 N is applied. Does it move?

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W = m g = 20 x 9.81 = 196.2 N
N = W = 196.2 N (level floor, no vertical load)
maximum static friction = 0.3 x 196.2 = 58.9 N
applied force 50 N < 58.9 N, so the block stays at rest
friction actually acting = 50 N
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The friction force only takes the value needed to hold the block, up to the maximum.